

Master's Thesis Presentation

QoS-Aware Element Allocation in Optical IRS-Aided HAP-Assisted Multi-UAV Networks

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Outline of Presentation

I. Introduction

II. Proposed Outage-Aware Spatial Resource Allocation

III. Numerical Results

IV. Conclusions

Outline of Presentation

I. Introduction

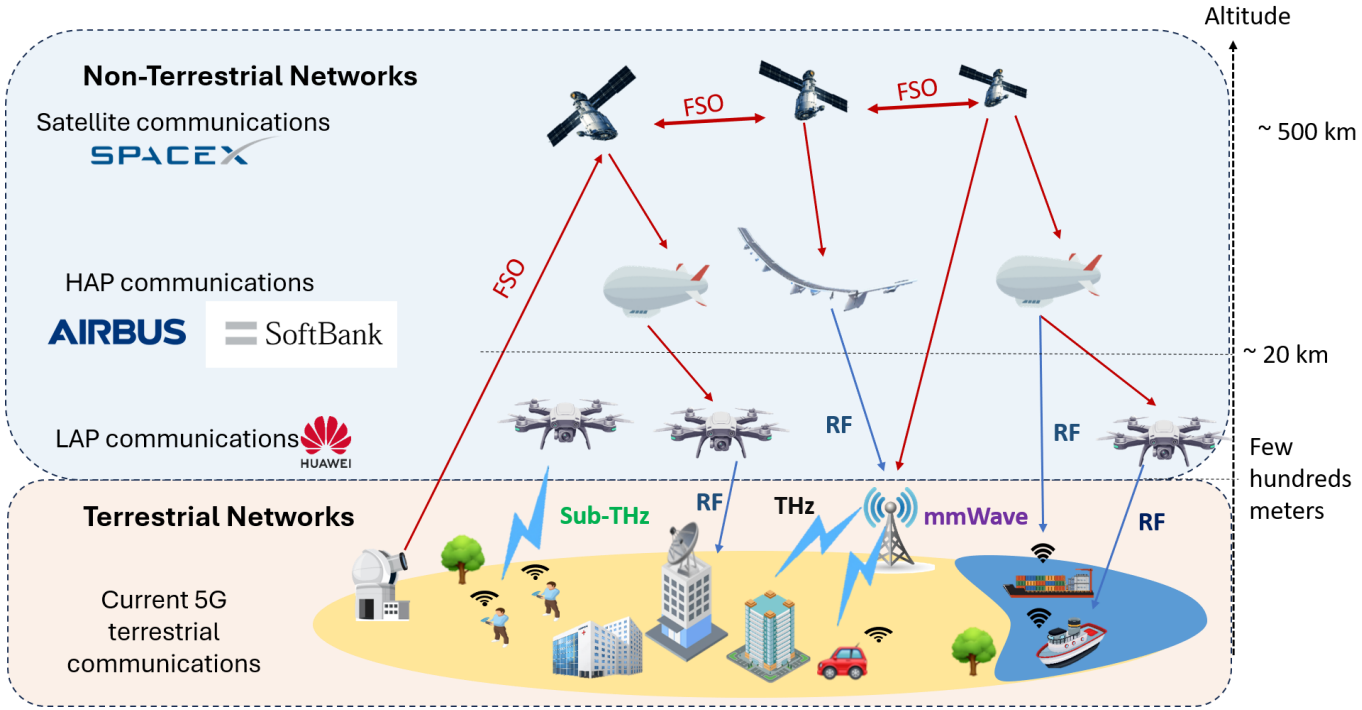
II. Proposed Outage- Aware Spatial Resource Allocation

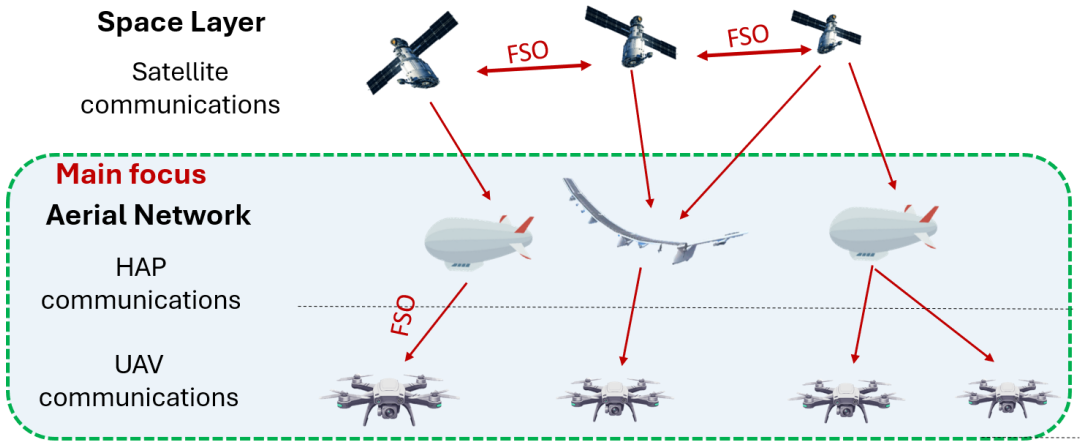
III. Numerical Results

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Aerial Networks for Wide Coverage

- Future 6G networks are expected to support *ubiquitous connectivity* and *high-rate services*, even in remote, rural, and hard-to-reach areas.
- Non-terrestrial networks (NTNs) *using satellites, HAPs and UAVs* can *extend service coverage beyond terrestrial infrastructure,* especially useful for connecting rural and remote areas.





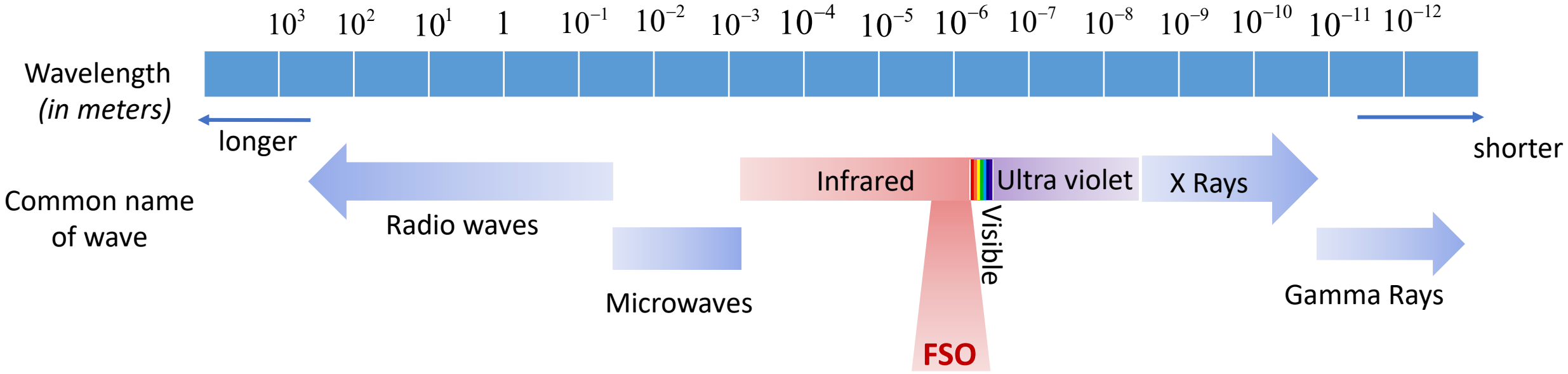
- Aerial networks with HAPs and UAVs:
  - Flexible deployment and wide coverage
  - Lower delay than satellites
  - Suitable for applications: temporary events, emergency, post-disaster communications

Aerial networks are promising components of wide-area future wireless networks.

High-Capacity Backhaul: FSO (1)

❑ Communications Technologies:

- ❖ **THz, sub-THz, mmWave: suitable for short-distance access links i.e., UAV to users**
- ❖ **Free space optical (FSO) communications: suitable for long-distance backhaul links, i.e., HAP to UAV**
  - ○ Uses narrow laser beams (near-infrared frequency bands *(200-400 THz))* to transmit data.
  - ○ Provides high-speed connections (Gbps and even Tbps)
  - ○ Operates in unlicensed optical spectrum



FSO-based aerial networks can provide high-capacity backhaul and wide coverage.

High-Capacity Backhaul: OIRS (2)

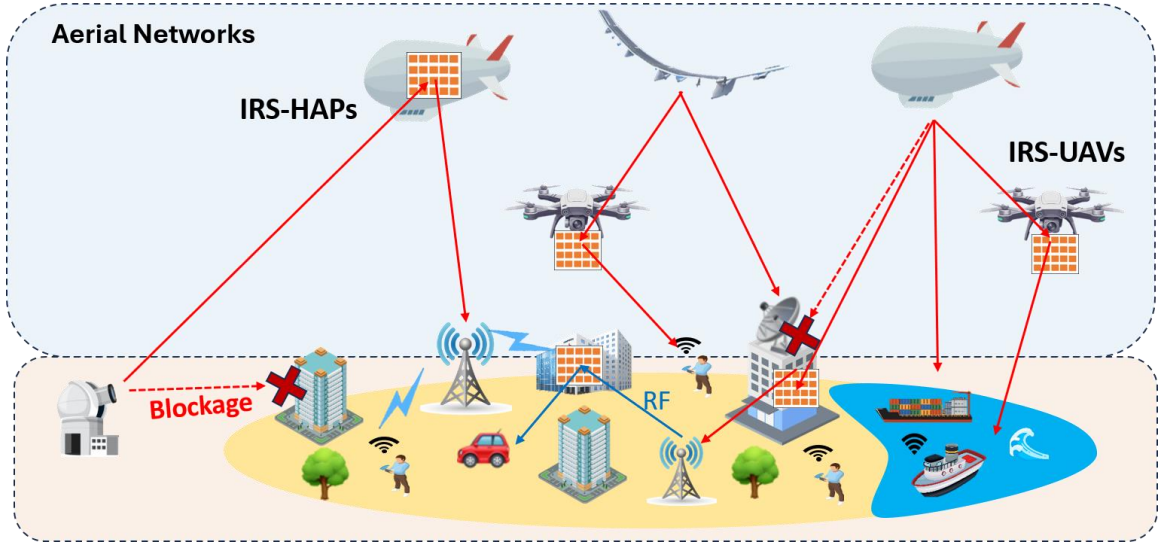
- FSO narrow beams require accurate alignment (strict light-of-sight), and blockage may interrupt direct FSO links
- Aerial platforms have strict *limits on cost, size, weight, and power consumption (C-SWaP) and hardware complexity*

Conventional relay approach:

- Uses optical receiver/transmitter modules
- *Increases payload, power consumption, and hardware complexity*
- Becomes *costly when supporting multiple UAVs* and increases the *burden to HAPs*

Optical Intelligent Reflecting Surface (OIRS)

- A passive surface: independent mirrors or meta-atoms
- *Reflects FSO beam* in a *controlled way*
- *One OIRS can support multiple UAVs by sharing OIRS elements*



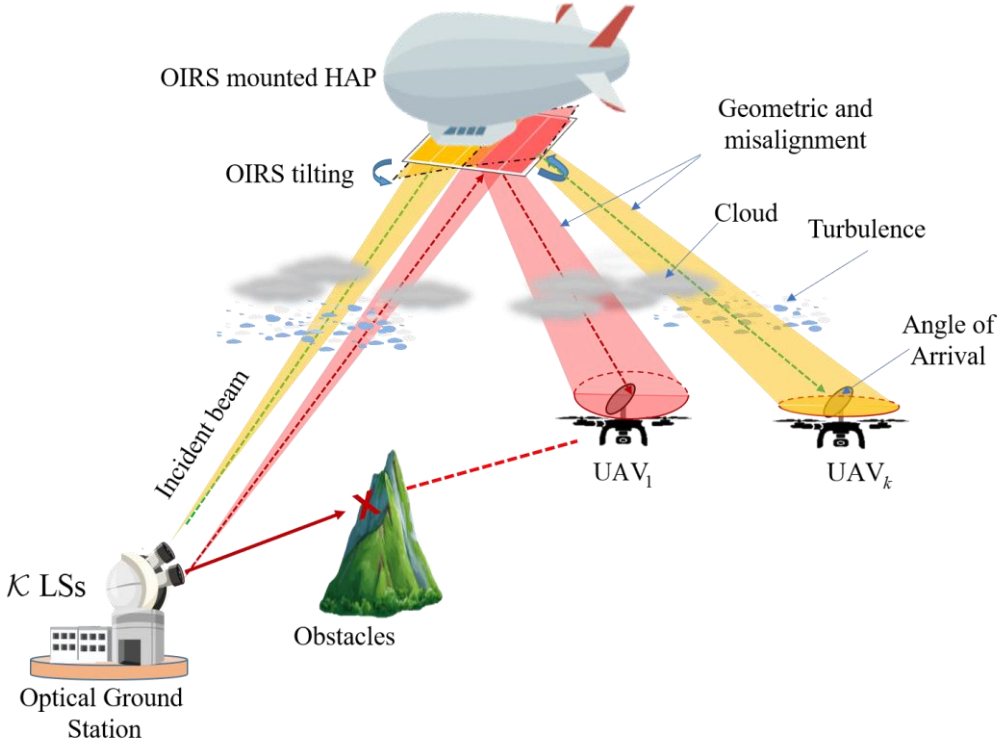
The concept of optical IRS mounted on aerial networks

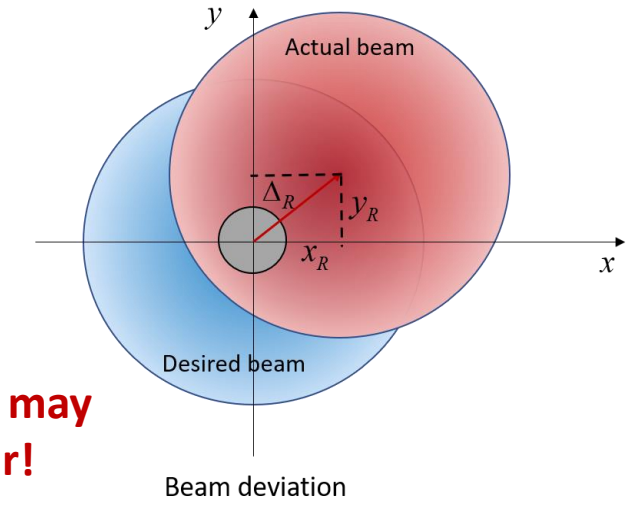
The OIRS mounted on HAP is a promising and efficient way to support multiple high-capacity UAV links.

Challenging Issues: FSO Impairments in Aerial Networks

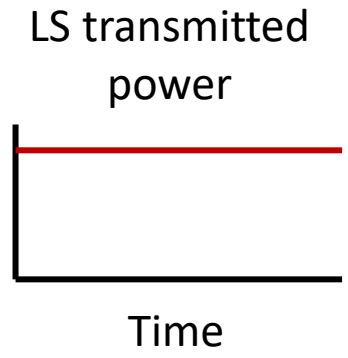
❖ **FSO links (LSs – HAP/OIRS – UAVs) are sensitive to**

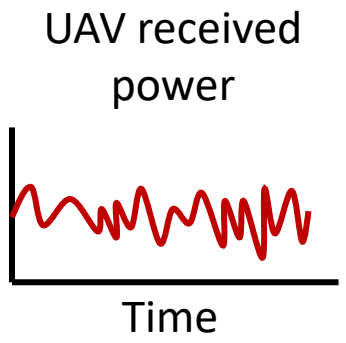
- **Cloud attenuation:** energy is absorbed by water droplets in clouds
- **Atmospheric turbulence:** received power fluctuates over time
- **Geometric and misalignment loss:** beam shifts and expands
- **Angle-of-arrival fluctuation:** received beam center may fall outside the UAV's field of view





Outage may occur!





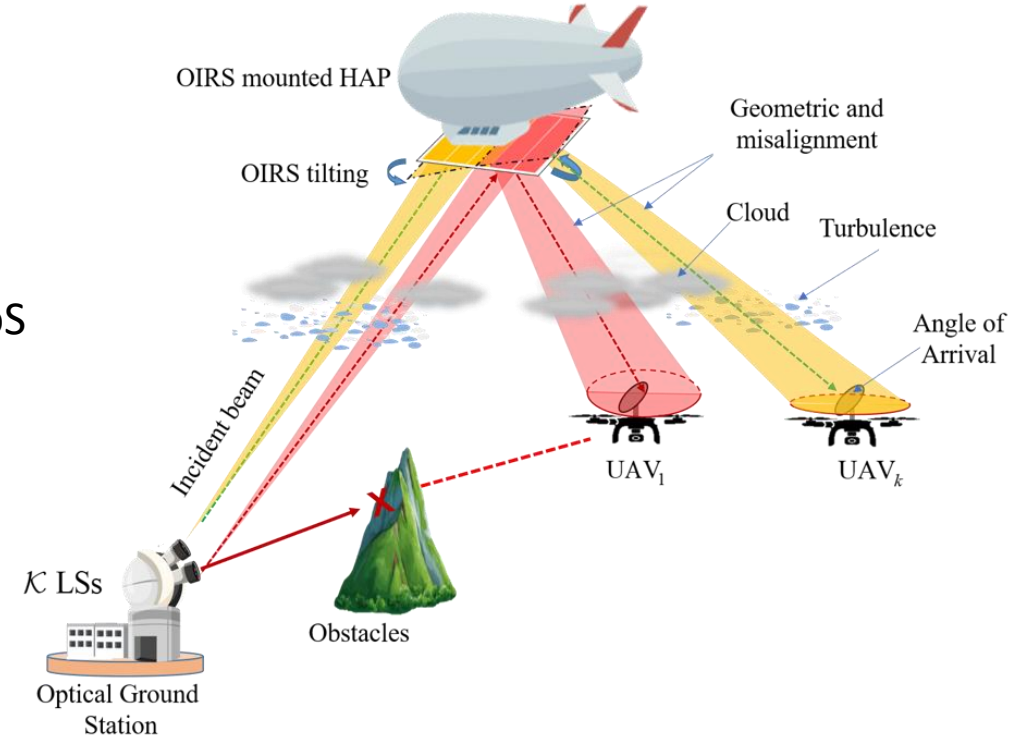
FSO impairments can degrade system performance and may cause link outage.

Problem Statement: OIRS Element Allocation under Heterogeneous FSO Channels

❖ Heterogeneous UAV links under FSO impairments:

- Different UAVs may experience different link qualities
- Each UAV must satisfy its Quality of Service (QoS)
- More OIRS elements can improve link quality, helping satisfy QoS

➡ *Each UAV may require a different number of OIRS elements to satisfy its QoS (outage probability)*



❖ However, OIRS elements are limited:

- Due to C-SWaP constraints on HAP, *OIRS elements* are *limited*
- *Multiple UAVs share the same OIRS*

➡ ***OIRS allocation is essential for supporting multiple UAVs while satisfying their QoS requirements.***

❑ Research question

➡ **How should limited OIRS elements be allocated among multiple UAVs under heterogeneous FSO channels?**

State of the Art on OIRS and Thesis Goals

1. OIRS channel modeling

- To the best of our knowledge, existing studies *mainly focus on OIRS channel modeling for single-user or terrestrial networks [1]-[5]*

  -> A comprehensive channel modeling for OIRS-based aerial networks has not been fully derived.

2. OIRS element allocation

- Few works *address OIRS allocation for multiple users*
  - *Simplified channel models / do not consider all FSO impairments [6]-[11]* ***-> need an accurate channel model***
  - *Based on equal or static allocation [4]-[6]* ***-> not optimize OIRS resources under heterogeneous channels***

❑ Our Goals/Contributions:

1. ***Accurate channel modeling:*** *this thesis derive* ***a comprehensive channel model*** *for OIRS-assisted FSO-based aerial networks, including cloud attenuation, atmospheric turbulence, geometric and misalignment loss, and UAV angle-of-arrival fluctuation.*

2. ***Efficient resource allocation:*** *this thesis develop an* ***OIRS element allocation scheme*** *that maximizes the number of QoS-satisfied UAV under heterogeneous channel conditions.*

Outline of Presentation

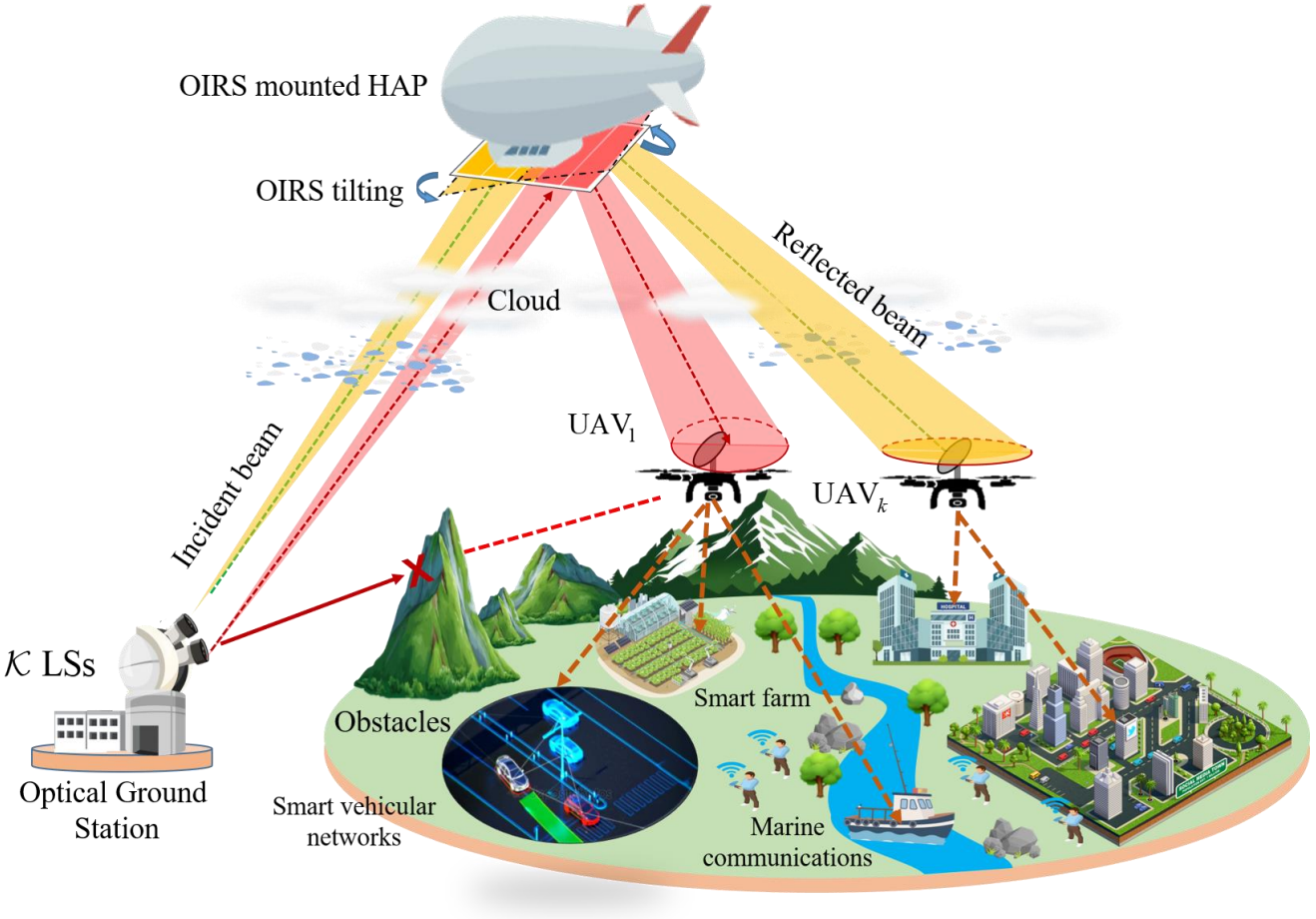
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System Model: OIRS-Aided HAP-Assisted Multi-UAV Network



The OIRS-aided HAP-assisted Multi-UAV network

1. Source: Ground Station

Multiple laser sources transmit FSO beams toward the HAP-OIRS

2. HAP-OIRS

The OIRS consists of multiple reflecting elements that redirect beams from the ground station toward UAVs

3. Destination: UAV

Multiple UAVs act as base stations, each with a QoS target

✓ *This thesis aim to allocate limited OIRS elements to multiple UAVs under heterogeneous channel conditions so that as many UAVs as possible can satisfy their QoS requirements*

Proposed Outage-Aware Spatial Resource Allocation (O-SRA)

- This thesis propose an **Outage-Aware Spatial Resource Allocation Scheme (O-SRA)** for OIRS to:

P1: Satisfy QoS (outage probability) for as many UAVs as possible

$$\mathcal{P}1: \max_{\{\chi_i, n_{x,i}\}} \sum_{i\in\mathcal{K}} \chi_i$$

Maximize number of QoS-satisfied UAVs

$$\text{s.t.} \quad P_{\text{out},i}(n_{x,i}) \le P_{\text{out},i}^{\text{target}}, \quad \forall i \in \mathcal{K},$$

Satisfy QoS (Outage probability)

Allocate OIRS arrays (OIRS elements)

$$\sum_{i\in\mathcal{K}} \chi_i n_{x,i} \le N_x,$$

Total allocated OIRS arrays <= total arrays

$$\chi_i \in \{0,1\}, \quad n_{x,i} \in \mathbb{Z}_+.$$

Active UAV index

- **After solving P1, if any OIRS arrays remain, we have a conditional problem P2**

✓ Allocate remaining OIRS arrays to improve total rate

$$\mathcal{P}2: \max_{\{n_{x,i}\}} \sum_{i\in S} R_i(n_{x,i})$$

Maximize the total rate of all supported UAVs

$$\text{s.t.} \quad n_{x,i} \ge n_{x,i}^{\star}, \quad \forall i \in S,$$

Final allocated OIRS arrays

$$\sum_{i\in S} n_{x,i} \le N_x,$$

Total allocated OIRS arrays <= total arrays

$$n_{x,i} \in \mathbb{Z}_+, \quad \forall i \in S.$$

P1: Maximize the Number of QoS-Satisfied UAVs (1)

- Before allocating OIRS arrays to UAVs, we need to calculate *how many OIRS arrays (elements) each UAV needs to satisfy its QoS*

$$y_i = h_i \times x_i + n$$

$$h_i = \eta \times h_{\text{cloud}} \times h_{\text{t}} \times h_{\text{gml}} \times h_{\text{AoA}}$$

Deterministic terms — *Random variables*

- PDF and CDF of the composite channel

$$f_h(h) = p_{\text{inFoV}} \frac{\xi^k \sigma_R^{k-1}}{h\sqrt{2\pi}} \exp\left(\frac{\sigma_R^2\xi^2}{4} + \frac{\xi A}{2} - \frac{A^2}{4\sigma_R^2}\right) D_{-k}(B) + \left(1 - p_{\text{inFoV}}\right)\delta(h), \quad (15)$$

where $A \triangleq \ln\left(\frac{h}{h_c G_{\text{gml}}}\right) + \frac{\sigma_R^2}{2}$, $B = \xi\sigma_R + \frac{A}{\sigma_R}$, *and* $p_{\text{inFoV}} \triangleq 1 - \exp\left(-\frac{\theta_{\text{FoV}}^2}{2\sigma_0^2}\right)$ *is the probability that the AoA* $\leq$ *UAV's FoV.*

$$F_h(h) \approx (1 - p_{\text{inFoV}}) + \frac{p_{\text{inFoV}}}{\Gamma(k)} \sum_{i=1}^{n} w_i^{(k)} \Phi\left(\frac{A_h + \frac{u_i^{(k)}}{\xi}}{\sigma_R}\right)$$

- Outage probability

$$P_{\text{out}} = (1 - p_{\text{inFoV}}) + p_{\text{inFoV}} \frac{p_{\text{inFoV}}}{\Gamma(k)} \sum_{i=1}^{n} w_i^{(k)} \Phi\left(\frac{A_\gamma(\gamma_{\text{th}}) + \frac{u_i^{(k)}}{\xi}}{\sigma_R}\right),$$

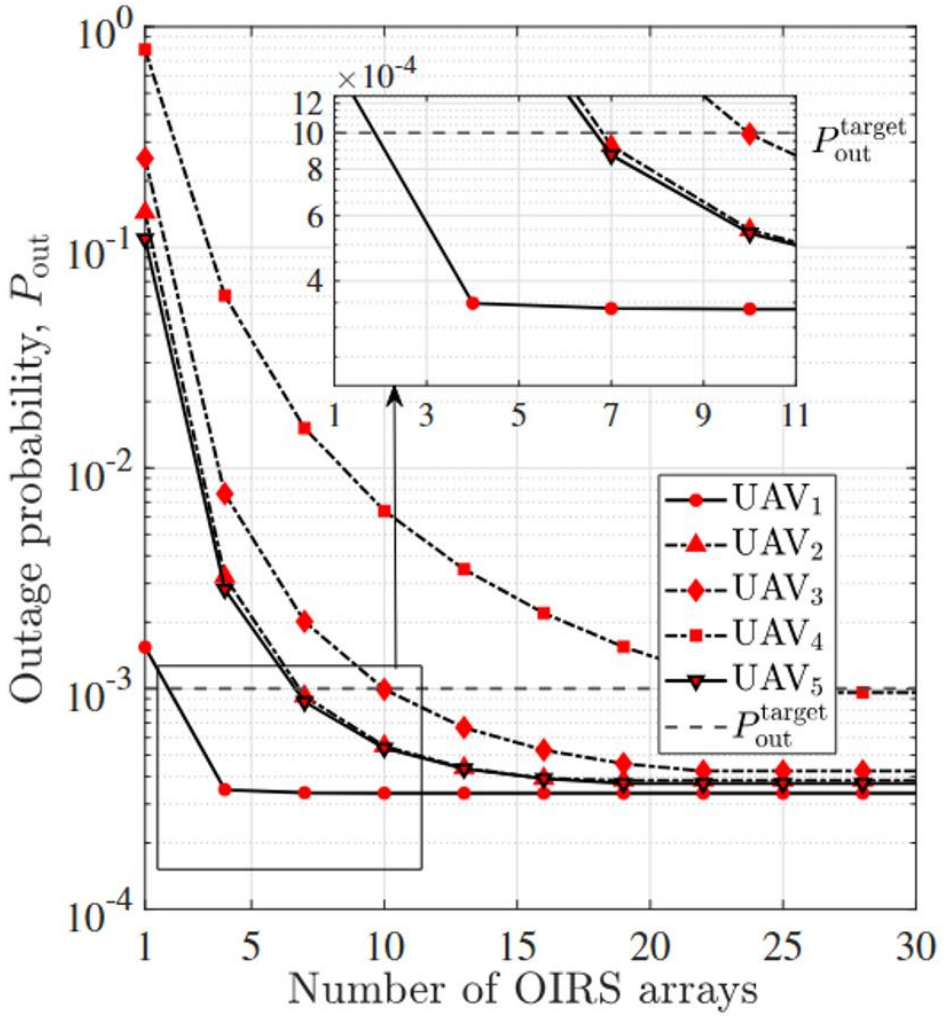


Figure 2. Outage probability over OIRS arrays

P1: Maximize the Number of QoS-Satisfied UAVs (2)

Determine required OIRS arrays for each UAV

Sort requirements in ascending order

Allocate OIRS arrays from the UAV with the lowest requirement to the highest requirement

**stop condition: all UAVs are supported or remaining arrays are not enough for the next UAV*

Maximum number of QoS-satisfied UAVs

Algorithm 1 UAV admission with Outage-Constrained QoS

Input: Total arrays N_x, outage targets $\{P_{\text{out},i}^{\text{target}}\}$, SNR thresholds $\{\gamma_{\text{th},i}\}$.
Initialize: $\mathbb{N} \leftarrow \emptyset$, $S \leftarrow \emptyset$.
for each UAV $i \in \mathcal{K}$ **do**
  Evaluate $P_{\text{out},i}(N_x)$ using (19).
  if $P_{\text{out},i}(N_x) > P_{\text{out},i}^{\text{target}}$ **then**
    continue (UAV i is infeasible).
  else
    Find continuous solution $L_{x,i}^{\star}$ of $P_{\text{out},i}(L_x) = P_{\text{out},i}^{\text{target}}$ using `fzero`.
    Map to array domain: $n_{x,i}^{\star} = \lceil L_{x,i}^{\star}/\Delta L_x \rceil$.
    Add $(i, n_{x,i}^{\star})$ to $\mathbb{N}$.
  end if
end for
Sort $\mathbb{N}$ in ascending order of $n_{x,i}^{\star}$.
$N_{\text{used}} \leftarrow 0$.
for each $(i, n_{x,i}^{\star})$ in sorted $\mathbb{N}$ **do**
  if $N_{\text{used}} + n_{x,i}^{\star} \leq N_x$ **then**
    $S \leftarrow S \cup \{i\}$.
    $N_{\text{used}} \leftarrow N_{\text{used}} + n_{x,i}^{\star}$.
  else
    break.
  end if
end for
$n_{\text{rm}} = N_x - N_{\text{used}}$.
Output: Supported UAV set S, required arrays $\{n_{x,i}^{\star}\}$, remaining arrays n_{rm}.

Conditional Problem P2: Improve Total Rate of QoS-Satisfied UAVs

- After P1, the admitted UAVs already satisfy QoS. If any OIRS arrays remain, we solve P2

Algorithm 2 Sum-Rate Maximizing Allocation in Phase II

1: **Input:** Admitted UAV set S; baseline arrays $\{n^{\star}_{x,i}\}$; remaining arrays n_{rm}.
2: **Initialization:** $n_{x,i} \leftarrow n^{\star}_{x,i}, \forall i \in S$.
3: **for** each UAV $i \in S$ **do**
4: Compute initial rate $R_i(n_{x,i})$ using (27).
5: **end for**
6: $R_{\mathrm{total}} \leftarrow \sum_{i \in S} R_i$.
7: **while** $n_{\mathrm{rm}} > 0$ **do**
8: **for** each UAV $i \in S$ **do**
9: Evaluate $\widetilde{R}_i = R_i(n_{x,i} + 1)$ using (27).
10: $\Delta R_i \leftarrow \widetilde{R}_i - R_i$.
11: **end for**
12: $i^{\star} \leftarrow \arg\max_{i \in S} \Delta R_i$.
13: $n_{x,i^{\star}} \leftarrow n_{x,i^{\star}} + 1$.
14: $R_{i^{\star}} \leftarrow \widetilde{R}_{i^{\star}}$.
15: $R_{\mathrm{total}} \leftarrow R_{\mathrm{total}} + \Delta R_{i^{\star}}$.
16: $n_{\mathrm{rm}} \leftarrow n_{\mathrm{rm}} - 1$.
17: **end while**
18: **Output:** Final allocation $\{n_{x,i}\}$ and total rate R_{total}.

After P1, remaining OIRS arrays

Allocate the remaining OIRS arrays one by one. We calculate the rate gain corresponding to allocating one additional array to each UAV

Choose the UAV with the highest marginal rate gain for each OIRS array

Maximize total rate

Key idea: incremental greedy algorithm

Complexity

$\mathcal{O}(M\, n_{\mathrm{rm}})$,

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Benchmarks and Parameter Settings

- **Benchmark for P1: UAV admission**
  - **Uniform Spatial Allocation (USA):** the OIRS is shared equally among all UAVs from the start.

- **Benchmarks for P2: remaining OIRS arrays optimization**
  - **Max-Gain Allocation (MGA):** aggressively allocate all remaining arrays to the UAV with the largest initial rate gain.
  - **Proportional Spatial Allocation (PSA):** arrays are allocated according to a ratio of the initial marginal rate gains.
  - **Particle Swarm Optimization (PSO):** a high-performance search-based benchmark.

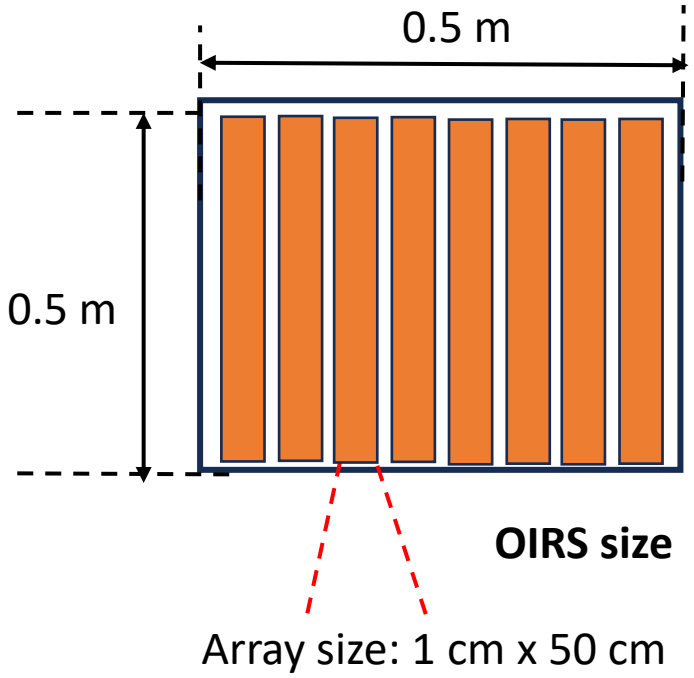


TABLE III: Simulation parameters for OIRS-Aided HAP-Assisted Multi-UAV System.

Name	Symbol	Value
OIRS-Aided HAP Parameters		
OIRS width along x-axis	L_x	0.5 m
OIRS width along y-axis	L_y	0.5 m
OIRS width resolution	ΔL_x	1 cm
HAP altitude	H_{HAP}	20 km
Reflection efficiency	η	0.95
Tilting variance	σ_0	10 μrad
Parameters of i-th UAV ($i = 1, 2, \ldots, M$)		
Number of UAVs	$\mathcal{K}$	5
UAV altitude	$H_{\mathrm{U},i}$	[100, 120, 150, 200, 180] m
UAV velocity	v_i	[15, 20, 25, 28, 30] km/h
Aperture radius	a	10 cm
Incidence angle	θ_i	[35, 24, 20, 26, 22]°
Wavelength	λ_i	[1550, 1570] nm
Cloud attenuation scale	$L_{c,i}$	[0.5, 1.3, 0.9, 1.5, 0.8]
Mean displacement	μ_x, μ_y	0
Hovering variance	σ_{ux}, σ_{uy}	0.2 m
Target outage probability (QoS)	P_{target}	10^{-3}

The Effectiveness of Proposed O-SRA

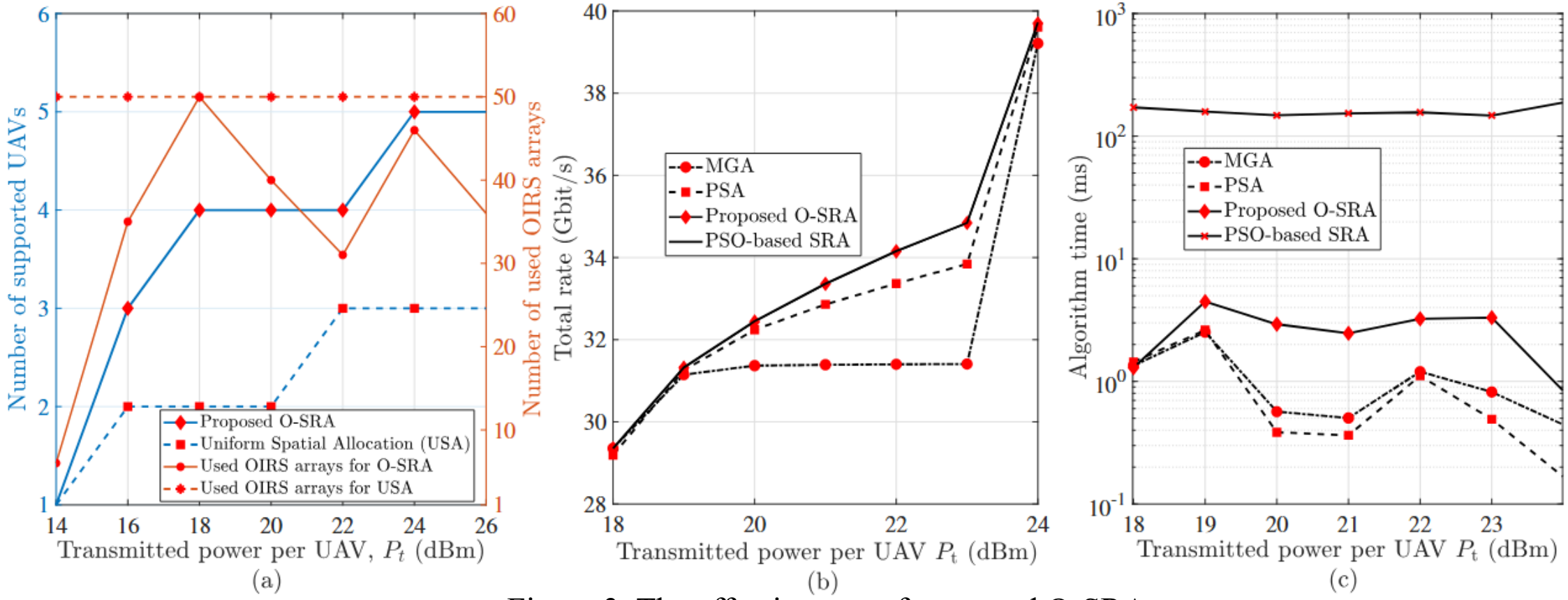


Figure 3. The effectiveness of proposed O-SRA.

- The proposed O-SRA ***outperforms the other allocation schemes*** by ***supporting more UAVs*** and providing a ***higher total achievable rate***. The total rate ***is close to the*** PSO benchmark while requiring ***much lower running time***.

- The proposed O-SRA ***ensures the minimum required OIRS arrays to as many UAVs as possible***, and then ***efficiently allocates the remaining arrays*** based on marginal rate gain.

Investigation of Proposed O-SRA in Mobility Scenario

- UAVs move away from HAP-OIRS for 2400 seconds at different velocities (Figure 4a)
- In Figure 4b, O-SRA ***consistently outperforms*** the two conventional schemes, demonstrating its effectiveness in adapting to dynamic UAV movements.
- O-SRA ***dynamically allocates*** the OIRS arrays for each UAV over time.

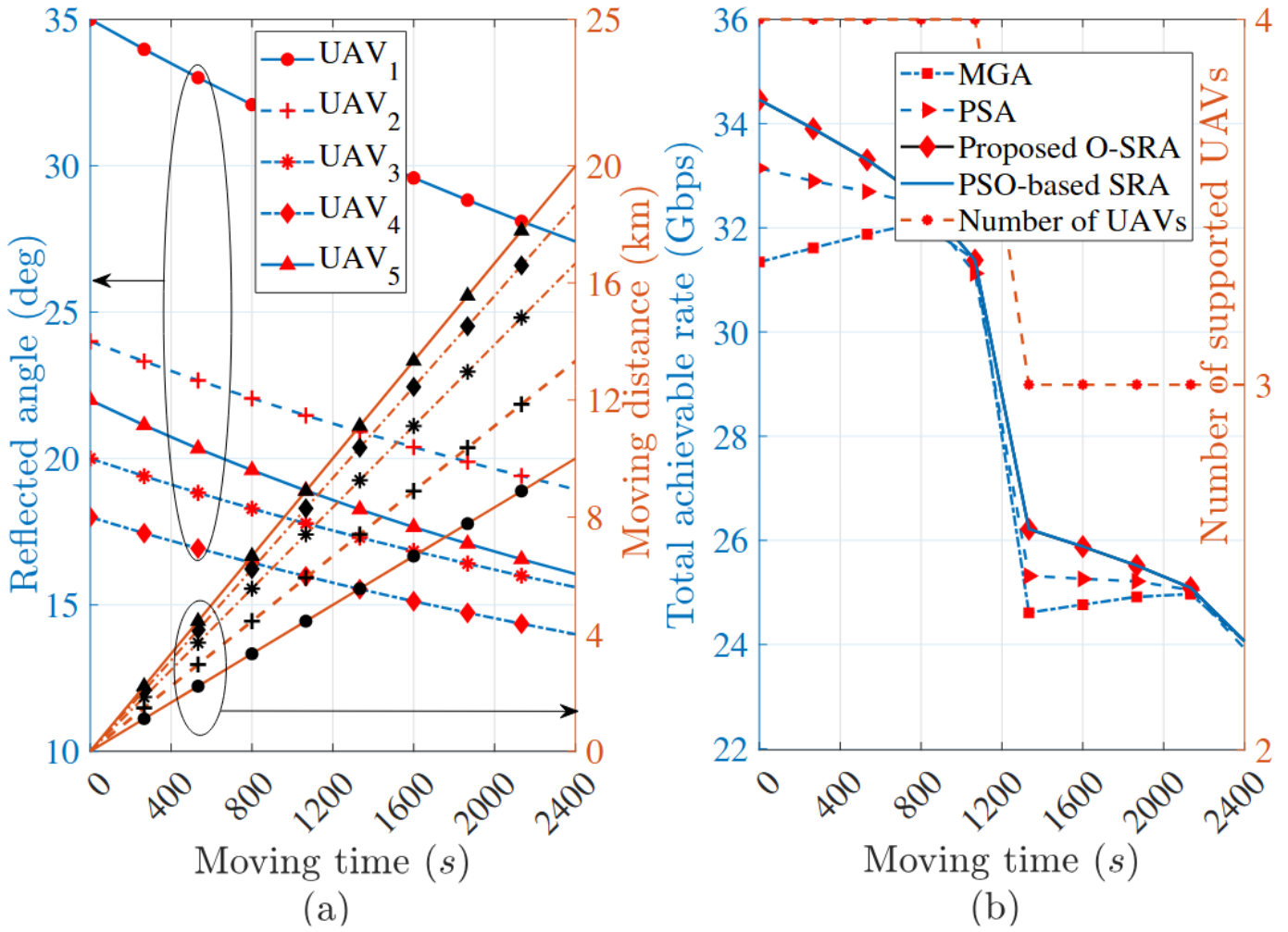


Figure 4. The effectiveness of proposed O-SRA in mobility scenario

Conclusions

- **This thesis proposed an *outage-aware spatial resource allocation* for optical IRS-aided HAP-assisted multi-UAV networks.**
  - The proposed scheme **effectively allocates** OIRS elements to **maximize the number of QoS-satisfied UAVs** first, and then **improves the total rate using the remaining OIRS resources**.
  - Numerical results confirmed the **effectiveness of the proposed scheme** compared to conventional schemes, including the **mobility scenario.**

- **Related Publications**

[J1] **Khanh D. Dang**, Hoang D. Le, Chuyen T. Nguyen, and Anh T. Pham, "QoS-Aware Element Allocation in Optical IRS-Aided HAP-Assisted Multi-UAV Networks," *IEEE Transactions on Communications*, May 2026 **(In Revision)**

[C1] **Khanh D. Dang**, Hoang D. Le, Chuyen T. Nguyen, Vuong V. Mai, and Anh T. Pham, "Spatial Resource Allocation for Optical IRS-Aided HAP-Assisted Multi-UAV Networks," *IEEE Vehicular Technology Conference,* Oslo, Norway, Jun. 2025

THANK YOU FOR LISTENING!

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[9] B.-M. Vu, N. T. Dang, and O.-S. Shin, “Optimization of mixed fso-rf downlink systems with slipt-enabled uav base station and hap-mounted optical ris,” IEEE Access, vol. 14, pp. 7099–7115, 2026.

[10] R. Deka, M. S. Alam, I. Ahmed, and S. Anees, “Performance analysis of a ris-haps assisted fso-uwoc system for ground-air-underwater connectivity,” in 2024 IEEE 100th VTC2024-Fall, 2024, pp. 1–5.

[11] H. D. Le, T. V. Nguyen, and A. T. Pham, “Aerial irs-aided vertical backhaul fso networks over fisher-snedecor f turbulence channels,” in2022 IEEE ICCE, 2022, pp. 133–138.

Appendix

A. Maximum-Gain Allocation (MGA)

The Maximum-Gain Allocation (MGA) assigns all remaining OIRS arrays to the UAV that yields the largest marginal rate gain at the Phase I operating point. Specifically, the marginal gain of UAV i is computed as

$$\Delta R_i^{\text{init}} = R_i(n_{x,i}^* + 1) - R_i(n_{x,i}^*).$$

The selected UAV index is

$$i^\star = \arg\max_{i \in S} \Delta R_i^{\text{init}},$$

and all remaining arrays are allocated to UAV $i^\star$:

$$n_{x,i^\star}^{\text{MGA}} = n_{x,i^\star}^* + n_{\text{rm}}, \qquad n_{x,j}^{\text{MGA}} = n_{x,j}^*, \ \forall j \neq i^\star.$$

Computational complexity: MGA requires evaluating one marginal rate increment for each UAV, resulting in $\mathcal{O}(M)$ rate computations and hence overall complexity $\mathcal{O}(M)$.

Key point:
MGA is aggressive. It gives all remaining resources to the UAV with the largest first-step rate gain.

B. Proportional Spatial Allocation (PSA)

The Proportional Spatial Allocation (PSA) distributes the remaining OIRS arrays proportionally to the marginal rate gains evaluated at the Phase I operating point. The marginal gain of UAV i is defined as

$$g_i = R_i(n_{x,i}^* + 1) - R_i(n_{x,i}^*).$$

The allocation weights are

$$w_i = \frac{g_i^+}{\sum_{j \in S} g_j^+}, \qquad g_i^+ = \max\{g_i, 0\},$$

and the final allocation is given by

$$n_{x,i}^{\text{PSA}} = n_{x,i}^* + \lfloor w_i \, n_{\text{rm}} \rfloor.$$

PSA performs only a single-shot marginal-gain evaluation and does not update the gains during allocation, which limits its ability to capture the nonlinear dependence of $R_i(n_x)$ on the OIRS size.

Computational complexity: PSA requires M rate evaluations to compute the marginal gains, thus the complexity is $\mathcal{O}(M)$.

Key point:
PSA is less aggressive than MGA.
It distributes remaining arrays according to marginal gains. However, it only uses the initial marginal gain once, so it does not update the gain after each allocation.

C. PSO-Based Spatial Allocation (PSO)

As a near-optimal benchmark, Particle Swarm Optimization (PSO) is applied to solve the discrete sum-rate maximization problem in Phase II:

$$\max_{\{n_{x,i}\}} \sum_{i \in S} R_i(n_{x,i})$$
$$\text{s.t.} \sum_{i \in S} n_{x,i} = n_{\text{rm}}, \ n_{x,i} \in \mathbb{Z}_+.$$

Each particle represents a candidate allocation vector $\mathbf{n}_x = [n_{x,1}, \ldots, n_{x,M}]$. The velocity and position of particle p at iteration t are updated according to the standard PSO rules [39].

Computational complexity: Let N_p denote the number of particles and N_{iter} the number of PSO iterations. Each iteration evaluates the objective function for all particles, and each evaluation requires computing M achievable rates. Therefore, the overall complexity is

$$\mathcal{O}(N_p \, N_{\text{iter}} \, M).$$

Each particle represents a split ratio of remaining arrays

$$\mathbf{w} = [w_1, w_2, ..., w_M], \quad w_i \geq 0, \quad \sum_i w_i = 1$$

$$n_i = n_i^* + n_{rm} w_i$$

Calculate total rate for each particle, remember the best value, update the global best, loop.

Appendix

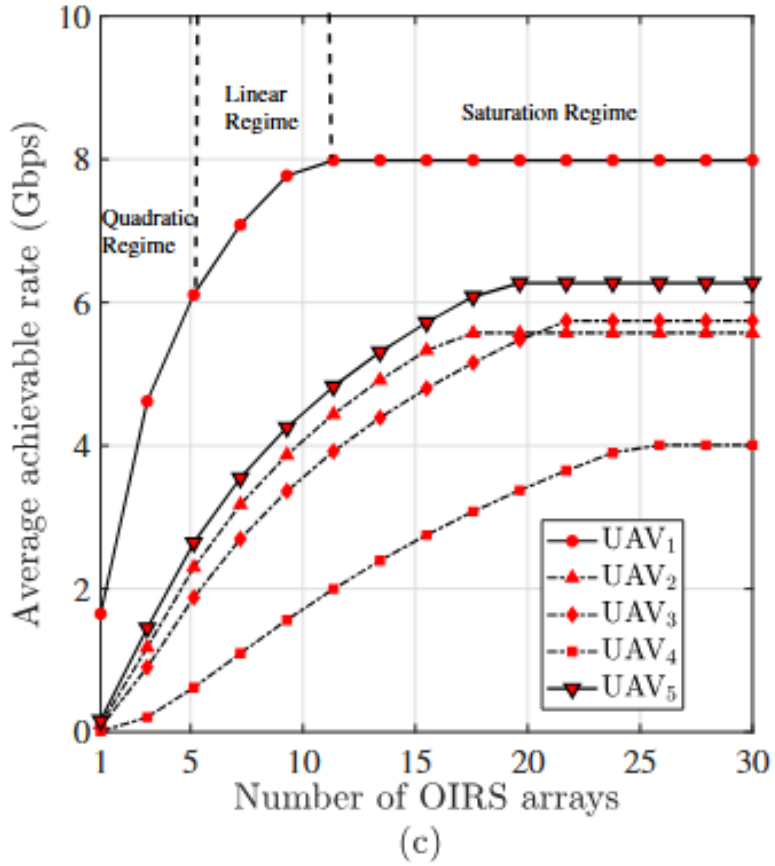
Average achievable data rate:

$$R \approx \frac{B\, p_{\text{inFoV}}}{\ln 2} \sum_{j=1}^{M} \frac{w_j}{1 - z_j} Q\left(\frac{\frac{1}{2} \ln\left(\frac{z_j}{(1-z_j)\Psi h_c^2 \Delta G^2} \right) + \frac{\sigma_R^2 \xi + 2k}{2\xi}}{\sqrt{\sigma_R^2 + \frac{k}{\xi^2}}} \right). \tag{22}$$

Update particles' velocities and positions:

$$\begin{aligned} \mathbf{v}_p(t+1) &= \omega \mathbf{v}_p(t) + c_1 \mathbf{r}_1(t) \odot \left(\mathbf{x}_p^{\text{best}} - \mathbf{x}_p(t) \right) \\ &\quad + c_2 \mathbf{r}_2(t) \odot \left(\mathbf{x}^{\text{gbest}} - \mathbf{x}_p(t) \right), \\ \mathbf{x}_p(t+1) &= \mathbf{x}_p(t) + \mathbf{v}_p(t+1), \end{aligned}$$

- Number of particles: $N_p = 30$
- Maximum iterations: $N_{\text{iter}} = 80$
- Inertia weight: $\omega = 0.7$
- Cognitive coefficient: $c_1 = 1.5$
- Social coefficient: $c_2 = 1.5$
- Random vectors: $r_1, r_2 \sim U[0, 1]$



(c)